

NANDINI SINGH

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SUMMARY

Data analytics professional with a strong academic foundation and hands-on project experience in data analysis, Machine Learning, and Deep Learning. Currently working on Generative AI concepts and practical applications. Skilled in SQL, Python, Excel, and data visualization, with a growing focus on applying analytics and AI techniques to solve real business problems. Demonstrated ability to analyze data, build insights, and support informed decision-making through structured, data-driven approaches

EXPERIENCE

AI/ML Research Intern	June 2025 – July 2025
Defence Research and Development Organisation (DRDO) – DEAL.	India, Dehradun
- Contributed to research on speech enhancement and noise reduction for real-world audio environments, focusing on improving audio quality under varying noise conditions. - Performed data collection, preprocessing, and feature extraction on audio datasets to support model training and experimentation. - Assisted in implementing and evaluating multiple machine learning and deep learning approaches, comparing performance across models and configurations. - Supported model evaluation and result analysis, documenting findings to aid research conclusions and further experimentation. - Collaborated with researchers to refine experimental pipelines and ensure reproducibility of results. Tools & Technologies: Python, TensorFlow, PyTorch, Librosa, NumPy, Kaggle.	

EDUCATION

Bennett University Greater Noida, India	2022 – 2026
Bachelor of Technology (B.Tech) in Computer Science & Engineering (Data Science)	CGPA: 8.2
Vantage Hall Girls' Residential School Dehradun, India	2019 – 2021
Class XII	81.7%
Class X	77%

PROJECTS

Adaptive Noise Cancellation using Deep Learning Python · Deep Learning · Kaggle
- Designed and implemented a speech enhancement system to reduce static and dynamic background noise in real-world audio signals as part of a research-oriented project. - Created a custom audio dataset by combining clean speech with multiple noise sources, and performed preprocessing and feature extraction using Librosa and NumPy. - Experimented with multiple deep learning architectures (CNN, LSTM, Autoencoders), identifying UNet as the most effective model. - Achieved best performance with UNet, reaching 49.32 dB PSNR, significantly improving speech clarity and noise suppression. - Evaluated model performance using quantitative metrics and qualitative waveform/spectrogram analysis.
End-to-End Customer Behavior Analysis Python · SQL · Power BI · Jupyter Notebook
- Conducted an end-to-end customer shopping behavior analysis to identify revenue drivers, customer segments, and subscription impact in a retail-style dataset. - Cleaned, processed, and explored data using Python (Jupyter Notebook), followed by KPI computation and business analysis using SQL. - Analyzed customer behavior patterns including high-value segments, repeat purchases, and subscription influence to support business decision-making. - Designed an interactive Power BI dashboard presenting KPIs, revenue trends, customer segmentation, and category performance for stakeholder-friendly insights.

- Translated analytical findings into **clear business insights and recommendations** to support marketing and product strategy decisions.

Temperature Forecasting using Machine Learning & Time Series | Python · Machine Learning

- Developed **time-series and machine learning models** to forecast city-level temperature trends using historical climate data.
- Performed **feature engineering** by extracting calendar-based features such as year, month, week, and day-of-week to improve model accuracy.
- Trained and evaluated **Linear Regression, Random Forest, and XGBoost** models to compare predictive performance.
- Achieved **strong predictive results with XGBoost**, reaching **R² scores up to 0.99**, outperforming baseline models.
- Visualized model performance using predicted vs actual plots, residual analysis, and feature importance charts to support interpretability.

YOLOv12 + DETR Hybrid Model for Object Detection (Ongoing Research) | *Python, TensorFlow, pycocotools, NumPy*

- Developing a hybrid model fusing predictions from YOLOv12M and DETR using WNMS and WBF.
- Performing evaluation on the COCO dataset to benchmark detection mAP and optimize hybrid model performance.

SKILLS

Programming Languages: Python, C++

SQL: Joins, Aggregations, Window Functions, Common Table Expressions (CTEs)

Libraries: Pandas, NumPy, Matplotlib, Seaborn

Data & ML Techniques: Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Machine Learning

Deep Learning Frameworks: TensorFlow, Keras, PyTorch

Visualization Tools: Power BI, Excel

Development & Analysis Tools: Jupyter Notebook, GitHub

Productivity Software: Microsoft Word, Microsoft PowerPoint

Platforms & Environments: Anaconda, Jupyter Notebook

Soft Skills: Analytical Thinking, Problem Solving, Clear Technical Communication, Structured Data Storytelling.

CERTIFICATIONS(*Linked on LinkedIn*)

- **Deloitte Data Analytics Job Simulation** — Forage — Certificate
- **Introduction to Artificial Intelligence** — Coursera — Certificate
- **Practical Time Series Analysis** — Coursera — Certificate
- **Supervised Machine Learning: Regression** — IBM — Certificate
- **Unsupervised Machine Learning** — IBM — Certificate
- **Neural Networks and Deep Learning** — Coursera — Certificate